Module 5: Arrays in Java

Java Laboratory – Problem Statements

Module Overview

Arrays are one of the most fundamental data structures in Java and serve as the foundation for advanced data structures such as lists, stacks, queues, trees, and graphs. This module introduces students to one-dimensional, two-dimensional, and jagged arrays while emphasizing real-world problem-solving through scenario-based programming exercises.

Instead of focusing solely on mathematical operations, the laboratory exercises simulate practical applications from education, banking, healthcare, retail, logistics, and business domains.

Total Practice Problems: 30

Difficulty Distribution

- Beginner: 10 Problems
- Intermediate: 12 Problems
- Advanced: 8 Problems

Learning Objectives

After completing this module, students will be able to:

- Declare and initialize arrays.
- Traverse arrays efficiently.
- Perform searching and sorting.
- Process large datasets.
- Develop matrix-based applications.
- Solve business problems using arrays.
- Build modular applications using arrays.

Concepts Covered

- One-Dimensional Arrays
- Two-Dimensional Arrays

- Jagged Arrays
- Array Traversal
- Searching
- Sorting
- Matrix Operations
- Statistical Calculations
- Data Analysis

Beginner Level Problems

Problem 1: Student Attendance Register

Difficulty: Beginner

Scenario

A faculty member wants to record the attendance status of students in a laboratory session.

Problem Statement

Develop a Java program to:

- Accept attendance (Present/Absent) for **30 students**.
- Count the number of students present.
- Count the number of students absent.
- Calculate the attendance percentage.
- Display an attendance summary.

Concepts Covered

- One-dimensional array
- Counting
- Traversal

Problem 2: Online Shopping Cart

Scenario

An online shopping portal allows a customer to purchase multiple products.

Problem Statement

Accept the prices of **10 products** purchased by a customer.

Display:

- Total Bill
- Highest-priced product
- Lowest-priced product
- Average product price

Problem 3: Monthly Temperature Analysis

Scenario

A weather department records the daily maximum temperature for one month.

Problem Statement

Store temperatures for **30 days**.

Display:

- Maximum temperature
- Minimum temperature
- Average temperature
- Number of days above average temperature

Problem 4: Student Marks Analysis

Scenario

A teacher wants to analyze the performance of students in a class test.

Problem Statement

Accept marks of **40 students**.

Display:

- Highest marks

- Lowest marks
- Class average
- Number of students who scored above average

Problem 5: Salary Report

Scenario

A company wants to analyze employee salaries.

Problem Statement

Accept salaries of **20 employees**.

Display:

- Highest salary
- Lowest salary
- Average salary
- Total salary expenditure

Problem 6: Mobile Recharge Tracker

Scenario

A telecom company records monthly recharge amounts of customers.

Problem Statement

Store recharge amounts for **15 customers**.

Display:

- Total recharge amount
- Average recharge
- Customer with highest recharge
- Customer with lowest recharge

Problem 7: Product Inventory

Scenario

A retail store wants to maintain stock information.

Problem Statement

Accept stock quantities of **25 products**.

Display:

- Products with zero stock
- Highest available stock
- Lowest available stock
- Total inventory

Problem 8: Daily Rainfall Report

Scenario

A meteorological department records rainfall every day.

Problem Statement

Accept rainfall data for **30 days**.

Display:

- Wettest day
- Driest day
- Average rainfall
- Number of rainy days

Problem 9: Cricket Scoreboard

Scenario

A cricket coach wants to analyze the team's performance.

Problem Statement

Store runs scored by **11 players**.

Display:

- Highest scorer
- Lowest scorer
- Team total
- Average score

Problem 10: Electricity Consumption

Scenario

An electricity provider wants to analyze household electricity consumption.

Problem Statement

Accept monthly electricity consumption for **12 months**.

Display:

- Maximum consumption
- Minimum consumption
- Annual consumption
- Average monthly consumption

Intermediate Level Problems

Problem 11: Banking Transaction Analyzer

Scenario

A bank wants to analyze daily transactions of an account.

Problem Statement

Store **30 daily transactions**.

Display:

- Total deposits
- Total withdrawals
- Largest deposit
- Largest withdrawal
- Net balance change

Problem 12: Hospital Patient Monitoring

Scenario

A hospital records patients' body temperatures.

Problem Statement

Store temperatures of **50 patients**.

Identify:

- Patients with fever
- Highest temperature
- Lowest temperature
- Average body temperature

Problem 13: Library Book Management

Scenario

A college library maintains the availability status of books.

Problem Statement

Store the availability status of **100 books**.

Display:

- Available books
- Issued books
- Total inventory

Problem 14: Employee Performance Dashboard

Scenario

A company records monthly performance ratings of employees.

Problem Statement

Accept ratings of **25 employees**.

Display:

- Highest rating
- Lowest rating
- Average rating
- Employees above average

Problem 15: Sales Performance Report

Scenario

A retail chain records monthly sales.

Problem Statement

Store sales data for **12 months**.

Display:

- Best sales month
- Worst sales month
- Average sales
- Growth compared to previous month

Problem 16: Online Examination Result Analysis

Scenario

An online examination portal stores marks of students.

Problem Statement

Display:

- Top 5 students
- Bottom 5 students
- Class average
- Pass percentage

Problem 17: Airline Seat Reservation

Scenario

An airline has **100 seats**.

Problem Statement

Maintain seat booking information.

Allow users to:

- Reserve a seat
- Cancel reservation
- Display available seats
- Display occupied seats

Problem 18: Parking Lot Management

Scenario

A shopping mall maintains parking slots.

Problem Statement

Track **50 parking spaces**.

Display:

- Occupied spaces

- Vacant spaces
- Occupancy percentage

Problem 19: Student Roll Number Search

Scenario

A university stores student roll numbers.

Problem Statement

Search a roll number using **Linear Search**.

Display whether the student exists.

Problem 20: Product Search

Scenario

An e-commerce platform stores product IDs.

Problem Statement

Search a product using **Binary Search** after sorting the IDs.

Problem 21: Sports Tournament Ranking

Scenario

A sports coordinator wants to prepare rankings.

Problem Statement

Store scores of participants.

Sort scores in:

- Ascending order

- Descending order

Display rankings.

Problem 22: Hospital Appointment Queue

Scenario

Store appointment IDs.

Sort them according to appointment number.

Display the sorted list.

Advanced Level Problems

Problem 23: College Result Processing (2D Array)

Scenario

A university stores marks of **100 students** in **6 subjects**.

Problem Statement

Calculate:

- Subject-wise average
- Student-wise total
- Percentage
- Highest scorer
- Topper

Problem 24: Bank Branch Cash Analysis (2D Array)

Scenario

A bank has **5 branches**.

Each branch records cash transactions for **7 days**.

Display:

- Branch-wise total
- Daily total
- Highest transaction day
- Overall average

Problem 25: Smart City Pollution Monitoring

Scenario

Different city zones record pollution levels every week.

Problem Statement

Store pollution data using a two-dimensional array.

Generate an analysis report.

Problem 26: Hospital Bed Management

Scenario

A hospital has multiple wards.

Each ward contains multiple beds.

Problem Statement

Represent the hospital using a **2D array**.

Display:

- Occupied beds
- Vacant beds
- Occupancy percentage

Problem 27: Cinema Seat Booking

Scenario

Represent a cinema hall using a **2D seating arrangement**.

Allow users to:

- Book seats
- Cancel bookings
- Display seating layout

Problem 28: School Timetable (Jagged Array)

Scenario

Different classes have different numbers of periods.

Problem Statement

Represent the timetable using a **jagged array**.

Display the complete timetable.

Problem 29: Sales Dashboard

Scenario

A company has multiple branches.

Each branch has different numbers of sales executives.

Problem Statement

Represent the data using a **jagged array**.

Generate branch-wise sales reports.

Problem 30: University Examination Analytics

Scenario

A university wants to generate a comprehensive examination report.

Problem Statement

Using arrays,

Generate:

- Department topper
- Subject topper
- University topper
- Pass percentage
- Grade distribution
- Rank list
- Subject-wise statistics

Challenge Problems

1. Develop a **Student Information Management System** using parallel arrays.
2. Design an **ATM Cash Inventory System** that tracks denomination counts.
3. Create a **Warehouse Inventory Management** application with stock updates and reorder alerts.
4. Build a **Weather Forecast Analyzer** that compares temperatures across multiple cities using a 2D array.
5. Implement a **Movie Ticket Booking System** with seat allocation, cancellation, and occupancy reports.
6. Develop a **Retail Sales Analytics Dashboard** that identifies top-performing products and branches.
7. Create a **Hospital Patient Monitoring System** that stores daily health readings and generates alerts.
8. Design a **School Examination Analytics System** that prepares subject-wise reports, grade distributions, toppers, and pass percentages.

Real-World Applications

Students will understand how arrays are used in:

- Banking transaction systems
- Student information systems
- Library management systems
- E-commerce applications
- Hospital management systems
- Airline reservation systems
- Cinema booking applications
- Smart city monitoring
- Weather forecasting
- Business analytics dashboards

Special Notes

- Encourage students to separate **data input**, **processing**, and **report generation** into different methods.
- Ask students to identify the appropriate array type (1D, 2D, or jagged) before writing code.
- Discuss the **time complexity** of traversal (`O(n)`), linear search (`O(n)`), binary search (`O(log n)` after sorting), and simple sorting algorithms ($O(n^2)$).
- Introduce students to software engineering practices by asking them to validate input, use meaningful variable names, and generate professional-looking reports.
- For advanced learners, encourage implementation using methods and classes, preparing them for the upcoming Object-Oriented Programming modules. This bridges the gap between procedural programming and object-oriented design.